Assignment #6

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Problem D.7, D.8

Find: (7a) Given open loop transfer func. from force to position where $z_e = 0$, find the open loop poles.

(7b) Using a modified PD control architecture (fig. 7-2), find the closed loop transfer function from z_r to z and the closed loop poles as a function of k_p and k_D.

(7c) Select k_p and k_D to place the closed loop poles at $p_1 = -1$ and $p_2 = -1.5$.

(7d) Using the gains from part (7c), implement the PD control in simulation and plot the response to a 1 m step input.

(8a) Given $t_r \approx 2$ seconds and $\zeta = 7.07$, find $\Delta_{ce}^{d}(s)$ and the associated poles. Find the corresponding k_p and k_D. Modify sim. to verify.

(8b) Given $F_{max} = 6$ N, modify the simulation. Use t_r as a tuning parameter so that the input just barely doesn't saturate.

Solution:

From D.5, the characteristic equation yields...

$$s^2 + \frac{b}{m}s + \frac{k}{m} = 0 \Rightarrow s = \frac{-\frac{b}{m} \pm \sqrt{\left(\frac{b}{m}\right)^2 - 4\left(\frac{k}{m}\right)}}{2} = \begin{cases} -0.05 - 0.773j \\ -0.05 + 0.773j \end{cases} \quad (7a)$$

Finding the T.F...

$$E \cdot k_p - k_D s Y = U$$

$$E = R - Y$$

$$Y = HU = H(E \cdot k_p - k_D s Y) = H((R-Y)k_p - k_D s Y) = H(Rk_p - Yk_p - k_D s Y)$$

$$= HRk_p - HYk_p - HYk_D s \Rightarrow Y + HYk_p + HYk_D s = HRk_p$$

$$\Rightarrow Y(1 + k_p H + k_D s H) = HRk_p \Rightarrow \frac{Y}{H} = \frac{Hk_p}{k_D s H + k_p H + 1} = \frac{k_p}{k_D s + k_p + \frac{1}{H}}$$

$$\Rightarrow \frac{Y}{H} = \frac{k_p}{k_D s + k_p + \frac{s^2 + a_1 s + a_0}{b_0}} = \frac{k_p b_0}{b_0 k_D s + b_0 k_p + s^2 + a_1 s + a_0} = \frac{k_p b_0}{s^2 + (a_1 + b_0 k_D)s + (b_0 k_p + a_0)}$$

And since for our particular case $b_0 = \frac{1}{m}$, $a_1 = \frac{b}{m}$, and $a_0 = \frac{k}{m}$, the poles are...

$$s = \frac{-(a_1 + b_0 k_D) \pm \sqrt{(a_1 + b_0 k_D)^2 - 4(b_0 k_p + a_0)}}{2} = \frac{-0.1k_D - 0.5 \pm \sqrt{k_D^2 + k_D - 20k_p - 59.75}}{10} \quad (7b)$$

Setting the characteristic equation coefficients equal...

$$(s-p_1)(s-p_2) = s^2 - p_1 s - p_2 s + p_1 p_2 = s^2 + (-p_1 - p_2)s + p_1 p_2$$
$$= s^2 + 2.5s + 1.5$$

$$\Rightarrow s^2 + 2.5s + 1.5 = s^2 + (a_1 + b_0 k_D) + (b_0 k_P + a_0)$$

$$\Rightarrow a_1 + b_0 k_D = 2.5 \Rightarrow k_D = \frac{2.5 - a_1}{b_0} = \frac{2.5 - \frac{b}{m}}{\frac{1}{m}} = \underline{12}$$

$$\Rightarrow b_0 k_P + a_0 = 1.5 \Rightarrow k_P = \frac{1.5 - \frac{k}{m}}{\frac{1}{m}} = \underline{4.5}$$ (7d)

From our plot we see that it doesn't even make it to the desired position.

Adjusting using t_r and ζ. First solve for ω_n...

$$s^2 + 2\zeta\omega_n s + \omega_n^2 = s^2 + 2\zeta\left(\frac{\pi}{2t_r\sqrt{1-\zeta^2}}\right)s + \left(\frac{\pi}{2t_r\sqrt{1-\zeta^2}}\right)^2$$
$$= s^2 + 1.57s + 1.233$$
$$\Rightarrow p = -0.7852 \pm 0.7854j$$ (8a)

And using similar calculation as above...

$$k_P = 3.17$$
$$k_D = 7.352$$

Using $F_{max} = 6\,N$, the maximum t_r is about 1.65 seconds (8b)

Problem E.8

Find:

(a) Using successive loop closure, draw a block diagram where $z_e = \frac{l}{2}$, input is z_r.

(b) For inner loop, find $k_{P\theta}$ and $k_{D\theta}$ for $t_{r\theta} = 0.25\,s$, $\zeta_\theta = 0.707$.

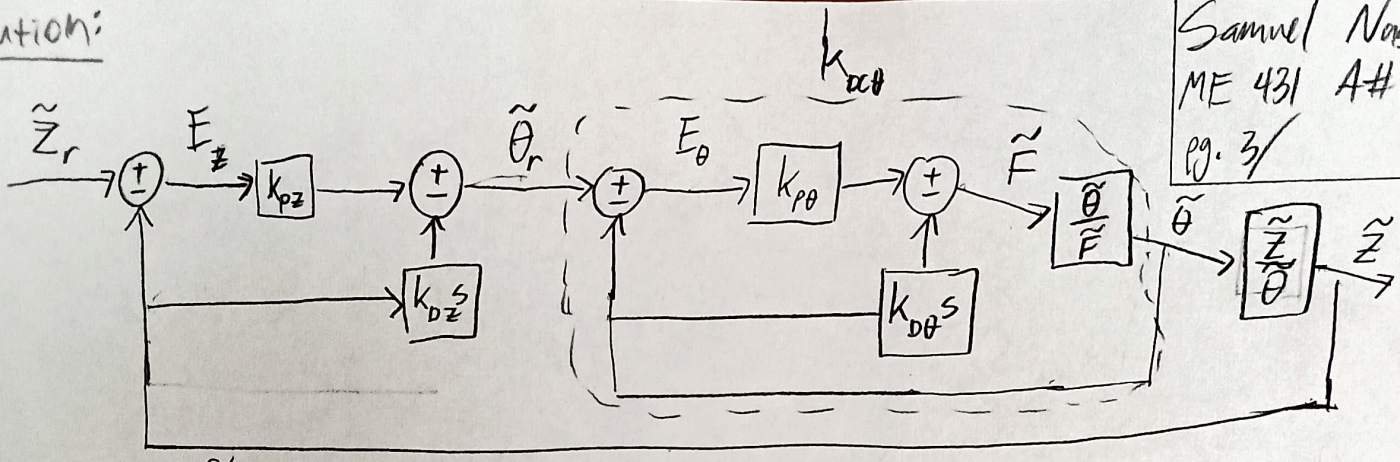
(c) Find DC gain $k_{DC\theta}$ for inner loop.

(d) Replacing inner loop w/ DC gain, find k_{Pz} and k_{Dz} for $t_{rz} = 10t_{r\theta}$ and $\zeta_z = 0.707$.

(e) Implement in simulation with command a square wave with $A = 0.25 \pm 0.5\,m$ and $f = 0.05\,Hz$. Use actual block position for feedback linearization for the equilibrium force.

(f) Modify sim for $F_{max} = 15\,N$. Tune t_{rz} to get fastest response w/o input saturation when $A = 0.25\,m$ for $\tilde{z}_r$.

ution:



(a)

Then for $\frac{\tilde{\theta}}{\tilde{F}}$ after simplifying assumption $m_1 g \tilde{z}$ goes to 0...

$$\frac{\tilde{\theta}}{\tilde{F}} = \frac{3l}{(m_2 l^2 + 3m_1 z_e^2)s^2} = \frac{3l}{(m_2 l^2 + 3m_1(\frac{l}{2})^2)s^2} = \frac{1}{(\frac{1}{3}m_2 + \frac{1}{4}m_1)l s^2}$$

$s = 0$

$$= \frac{\frac{1}{(\frac{1}{3}m_2 + \frac{1}{4}m_1)l}}{s^2} \Rightarrow \begin{aligned} a_0 &= 0 \\ a_1 &= 0 \\ b_0 &= \frac{1}{(\frac{1}{3}m_2 + \frac{1}{4}m_1)l} \end{aligned}$$

Then we have...

$$\Delta_{cl}^d = s^2 + (a_1 + b_0 k_D)s + (a_0 + b_0 k_p) = s^2 + 2\zeta\omega_n s + \omega_n^2$$

Given $t_{r\theta}$ and ζ_θ...

$$\omega_n = \frac{\pi}{2t_r\sqrt{1-\zeta^2}}$$

Then setting coefficents equal yields $k_{p\theta} = 46.044$ and $k_{D\theta} = 7.328$ (b)

Then the DC gain is...

$$k_{DC\theta} = \lim_{s\to 0} \frac{b_0 k_p}{s^2 + (a_1 + b_0 k_D)s + (a_0 + b_0 k_p)} = \frac{b_0 k_p}{a_0 + b_0 k_p} = 1$$ (c)

Then for the outer loop t.f. with the gain...

$$k'_{DC\theta}\frac{\tilde{Z}}{\tilde{\theta}} = \frac{-k_{DC\theta} g}{s^2} \Rightarrow \begin{aligned} b_0 &= -k_{DC\theta} g \\ a_0 &= 0 \\ a_1 &= 0 \end{aligned}$$

$$\omega_n = \frac{\pi}{2(10t_{r\theta})\sqrt{1-\zeta^2}}$$

Then by same logic we have $k_{pz} = -0.081$ and $k_{Dz} = -0.128$